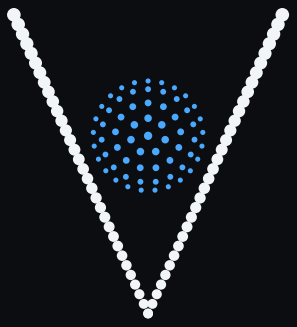
NASA SPACE APPS CAIRO 2026

Challenge: Create Health Monitoring Software for Astronauts on Space Missions



When Earth is nineteen light-minutes away, the crew is its own first responder.

A health system that watches, escalates, guides and remembers, onboard, with no link home.

VitalOrbit

Autonomous onboard crew health for deep-space missions

Team CUPIQ · Egyptian Chinese University · Cairo

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AT A GLANCE

What it is	An onboard health system for a Mars crew: it detects, guides, forecasts and remembers, without waiting for Earth.
Built	Python, FastAPI, scikit-learn, SQLite and a no-build React front end. One process, one command, no cloud.
Data	Real Fitbit wearable streams from four people (CC-BY 4.0) for heart rate, activity and sleep; simulated SpO2, temperature and cabin. Every number tagged REAL, SIM or MODEL.
Measured	3,456 readings scored, 12 of 12 injected emergencies detected, 45 min median to first alert, 0 false critical alerts, 0.08 false warnings per crew-day.
Mission modelled	Day 432 of a 920-day Mars mission, crew on the surface, 19.1 light-minutes from Earth, 520.4 mSv accumulated.
Repository	github.com/3AMURR/VitalOrbit

SUBMISSION PACKAGE

Item	What we submit
Public repository	github.com/3AMURR/VitalOrbit, with code, data, documents and a one-command start
30-second video	The working product in thirty seconds, narrated
Presentation video	The seven-slide story explained, 2:17
This document	Problem, solution, evidence, limitations, and how to reproduce the demo

1 The problem

On a Mars mission the doctor is not late. The doctor is nineteen light-minutes away, and the answer takes nineteen minutes to come back.

Today astronaut health depends on flight surgeons at Mission Control. That works on the International Space Station, where the crew can reach the ground within seconds and a vehicle can come home in hours. Neither condition holds on the way to Mars or on its surface. A question and its answer cost 38 minutes at our mission point, and during solar conjunction, when the Sun sits between Earth and the ship, the link disappears for about two weeks with no workaround.

What makes this more than a communications problem is the hazard profile. NASA's Human Research Program organises the risks of spaceflight into five categories, RIDGE: **Radiation**, **Isolation** and confinement, **Distance** from Earth, **Gravity** fields, and the hostile, closed **Environment**. On a long mission these do not announce themselves. Bone and muscle are lost quietly, the optic nerve swells, dose accumulates against a career limit, mood erodes under confinement, and cabin CO2 drifts. By the time a symptom is obvious the intervention window has often closed, and the person who would have noticed the trend is on the wrong planet.

So the crew is its own first responder. What it lacks is not knowledge, which can be carried onboard, but continuity of attention: something that knows each astronaut's normal, notices a deviation while it is still small, tells them what to do next, and remembers the whole mission when a human memory would not.

2 The solution

VitalOrbit is one onboard system with four responsibilities that normally sit with four different people on the ground.

It watches. One anomaly model per astronaut, trained only on that person's own fourteen days of heart rate, activity and sleep, scores every five-minute reading against what is normal for them at that hour and that activity level. Exercise is not an anomaly; a resting heart rate fifty beats above their own baseline is. Clinical limits run alongside the model as a safety net, so a value that is dangerous but statistically familiar still raises a flag.

It escalates, and it asks. A single odd reading is noise. An alert opens only when five of the last six readings are abnormal, and then escalates on persistence: 30 minutes, 90 minutes, three hours. At the thirty-minute mark it does something no monitor does, which is to ask the astronaut how they feel. "I was exercising" closes the alert and labels it a false positive, which is kept as feedback. "I don't feel well" opens the assistant on the right protocol.

It guides. The assistant is a real chatbot with three brains in fallback order: Claude when a link to Earth exists, a local model through Ollama when it does not, and deterministic onboard protocol cards when neither is available. The safety-critical content never comes from the language model: steps, timers and red flags are retrieved from the onboard knowledge base, every drug it names is checked against pharmacy stock and radiation-adjusted expiry, and dispensing needs a confirmation and is logged.

It remembers. Long-mission risks are projected to mission end and to the day each crosses its limit: radiation against the 600 mSv career limit, bone and muscle loss from each person's real exercise minutes, SANS, behavioural health from daily check-ins and real sleep, cabin CO2. Case reports to

Earth are assembled as SOAP notes and travel at light speed, arriving stamped with the delay that produced them.

3 What is actually built

All of it runs today. The repository holds 3,055 lines of application code: a 470-line simulation and detection engine, a 430-line API of roughly thirty endpoints, a 360-line assistant, the RIDGE risk model, a SQLite pharmacy, and a React front end with no build step. `./run.sh` installs dependencies into a virtual environment and serves the whole application on port 8000.

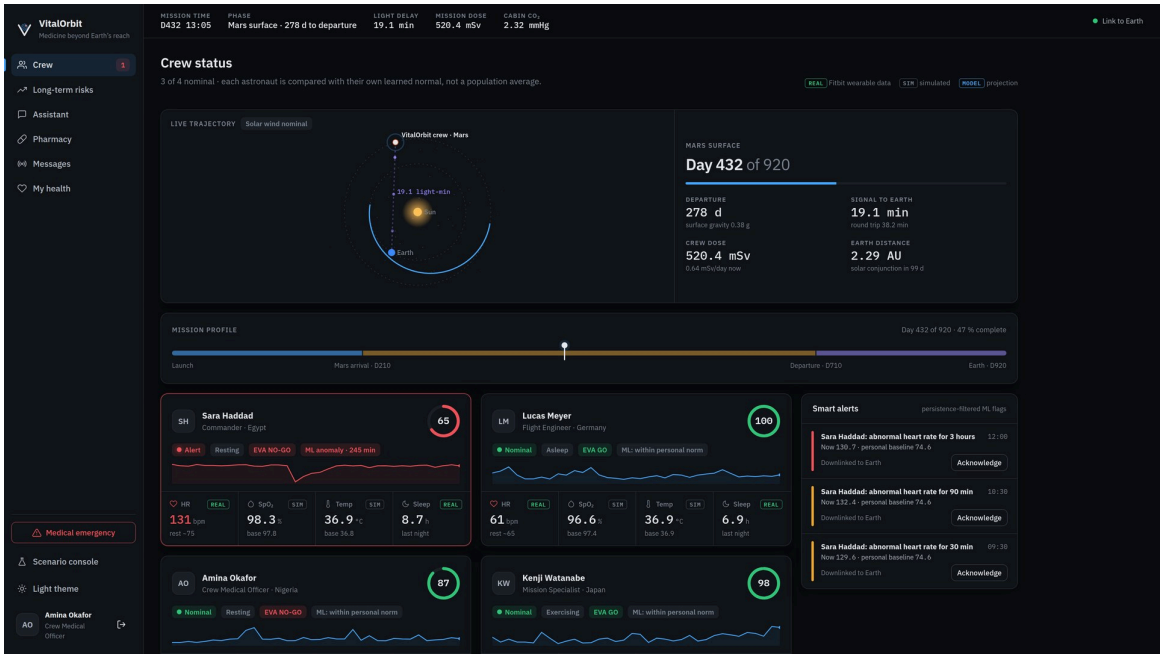


Figure 1: Crew status as the Crew Medical Officer sees it. The trajectory, light delay and dose are computed from orbital geometry, not typed in. Sara Haddad's card is red because a live anomaly is in progress: 131 bpm against a personal resting baseline of 75, with the alert column showing escalation at 30 minutes, 90 minutes and 3 hours.



Figure 2: One astronaut. The health score is 100 minus visible penalties, so every point lost can be traced to a vital. Real wearable data drives the heart-rate and sleep traces; SpO2 and temperature are simulated and carry a SIM tag. Below, the risk cards show 520 mSv against the 600 mSv career limit and the day each limit is crossed.

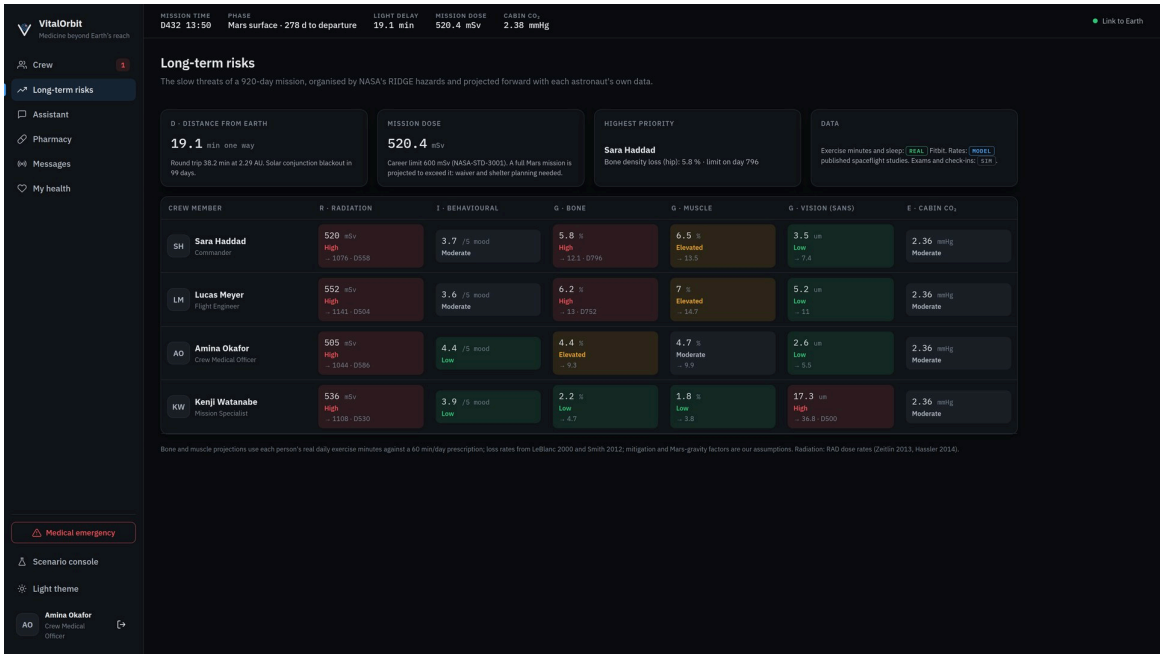


Figure 3: The long-mission board, organised by the five RIDGE hazards. Each cell carries the current value, its level, the projection to mission end and the day it crosses the limit. Bone and muscle projections use each astronaut's real daily exercise minutes against a 60-minute prescription.

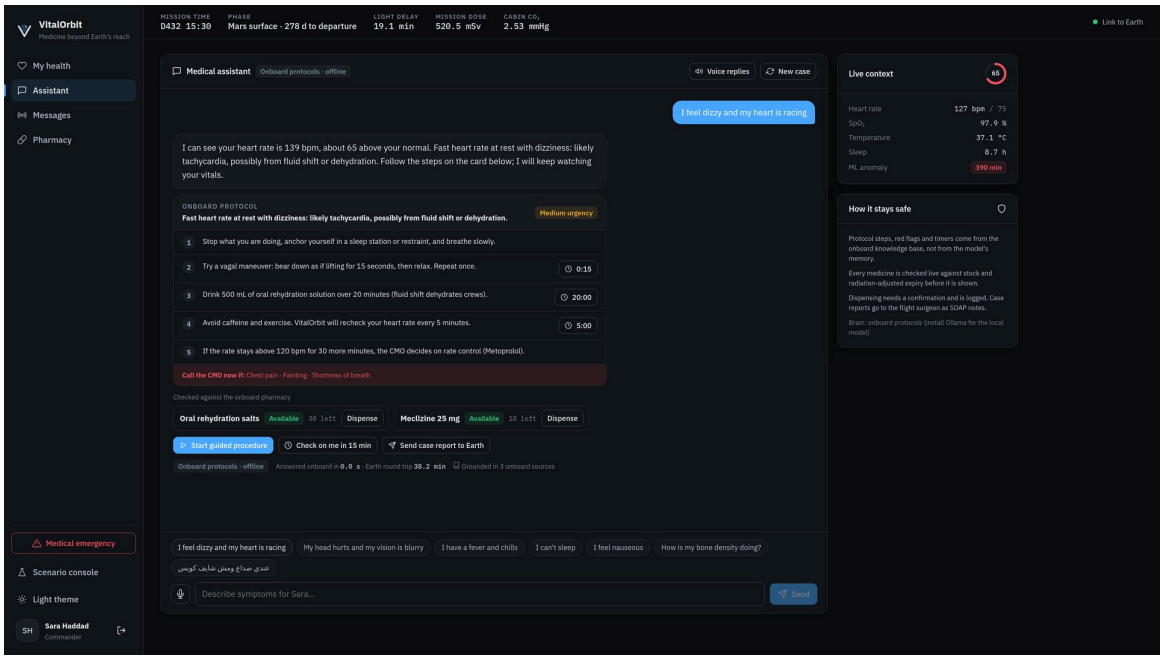


Figure 4: The assistant answering offline, from onboard protocols, after a vague complaint. It has already read the sensors, so it knows which protocol applies; the steps carry timers, the drugs are checked against stock and effective expiry, and the case report to Earth is one button away.

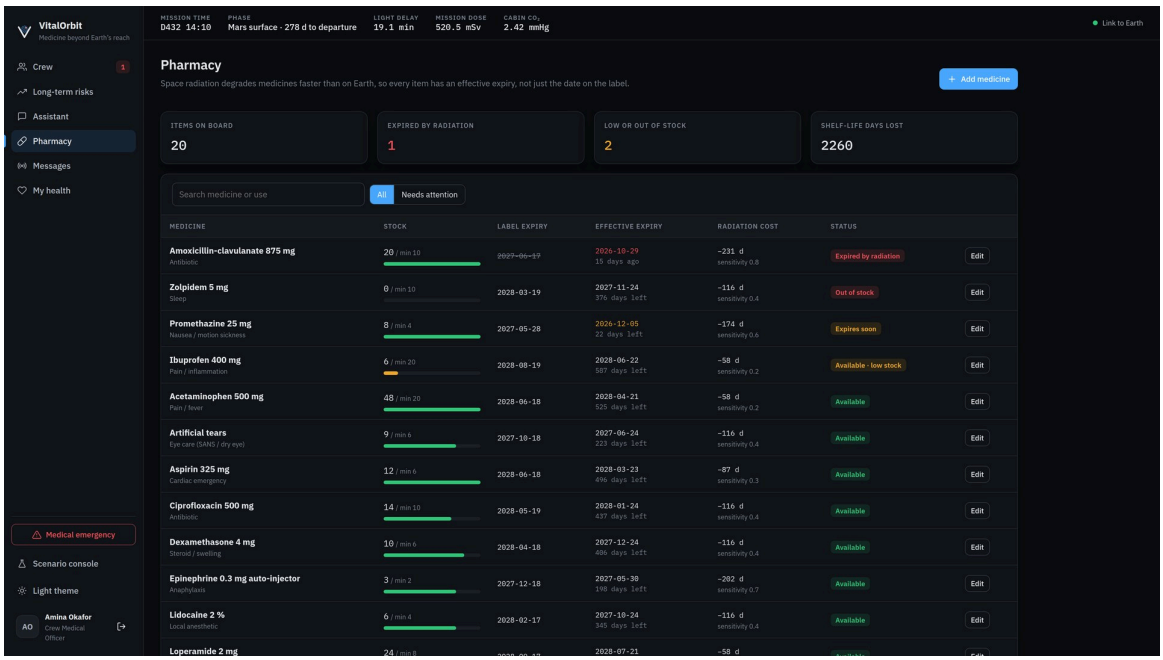


Figure 5: The pharmacy. Amoxicillin-clavulanate is still valid by its label date but has expired under radiation, with 231 days of shelf life lost; 2,260 days have been lost across the inventory.

4 Data, and what we can honestly claim

The weakest point of most hackathon health projects is the data, so we checked ours before building on it. NASA's Open Science Data Repository holds no public astronaut vital-sign time series: it is overwhelmingly omics data, and the human medical data that exists is controlled-access and takes weeks to obtain. Simulating vitals and calling them astronaut data would have been the easy path and the dishonest one.

Instead, heart rate, activity and sleep are real: five-minute streams from four participants in the public Crowd-sourced Fitbit dataset (Furberg et al., Zenodo 53894, CC-BY 4.0), replayed day by day as the four astronauts. SpO2, skin temperature, cabin environment, SANS exams and the check-in history are simulated, because a wrist wearable does not measure them. Every number in the interface is tagged REAL, SIM or MODEL, and the API returns the basis for each risk projection.

Source	What it grounds
Fitbit wearable dataset, Zenodo 53894 (CC-BY 4.0)	Heart rate, activity and sleep for four people; training data for every personal model
RAD dose rates: Zeitlin et al. 2013, Hassler et al. 2014	1.8 mSv/day in transit and 0.64 mSv/day on the surface
NASA-STD-3001 Vol. 1 Rev. B	The 600 mSv career effective dose limit
LeBlanc et al. 2000; Smith et al. 2012	Bone loss rates and the effect of exercise countermeasures
Mader et al. 2011	SANS prevalence and the optic-disc findings we simulate
Du et al., AAPS Journal 2011	Drug instability in flight, the basis of the radiation-adjusted expiry
NASA HRP hazards and risks	The RIDGE structure of the risk board

> Our own assumptions are labelled as such in the code and in the interface: per-drug radiation sensitivity, the Mars-gravity factor that slows bone loss, the exercise mitigation curves, and the simulated SANS exams and check-in history. We would rather a judge find them declared than find them.

4.1 Does the detection work?

Yes, and the number is reproducible: `scripts/evaluate_ml.py` runs the real engine over three normal mission days and then injects twelve emergencies, writing its results to `data/ml_evaluation.json`.

Measurement	Result
Readings scored (4 crew × 3 days × 288)	3,456
Injected emergencies (tachycardia, hypoxia, fever × 4 crew)	12
Detected, with the correct vital named	12 of 12
Median time to first alert	45 min
Escalated to critical within 3 hours	12 of 12
False warnings on normal days	1 (0.08 per crew-day)
False critical alerts	0

The interesting number is not the twelve of twelve. It is the forty-five minutes: the system does not wait for a clinical threshold to be crossed, it notices the deviation while the astronaut still feels roughly fine, which on a Mars mission is the only window in which the answer is still cheap.

5 Relevance to the challenge

The challenge asks for health monitoring software for astronauts on space missions. Our reading is that the hard part is not the dashboard; it is that the monitoring must survive the absence of the people who normally interpret it.

What the mission needs	What VitalOrbit does
Continuous monitoring with no ground in the loop	Per-astronaut anomaly detection that runs onboard and needs no network
Early warning rather than late confirmation	Persistence-based escalation from 30 minutes, naming the vital involved
Immediate guidance for a crew with no doctor	Offline protocols with timers, drug checks against real stock, guided procedure mode
Care for slow hazards, not only emergencies	The five RIDGE hazards projected to mission end, with the day each limit is crossed
Communication across a 38-minute round trip	Store-and-forward messaging, light-speed delay applied to the data, SOAP case reports, conjunction queueing
Privacy and medical authority	Roles enforced in the API: an astronaut cannot read a crewmate's record; the flight surgeon is delayed, not privileged to real time
A crew that must trust the system	Every alert shows its baseline, every score shows its penalties, every number shows its provenance

6 Impact

The immediate beneficiaries are four to six people who cannot be evacuated, cannot be operated on from the ground, and cannot wait. The measurable difference is time: a deviation is surfaced at 45 minutes instead of at the next scheduled check-in, and a protocol is in front of the astronaut within seconds of saying they feel unwell, instead of after a 38-minute round trip and a flight surgeon's queue. The second beneficiary is the flight surgeon, who receives structured SOAP notes with the relevant history attached instead of a voice loop and a guess, and can spend the light-time delay deciding rather than reconstructing.

Beyond Mars the same structure applies wherever the doctor is far away: Antarctic stations through the winter, submarines, offshore rigs, remote clinics and field hospitals. The models are personal and the constraints are generic, which is the shape of the problem in those settings too.

7 Originality

Four choices separate this from a monitoring dashboard.

The model is personal, not population-based. A fixed threshold on heart rate would flag every astronaut during exercise and miss the fit one at rest. Comparing each reading to that person's own rhythm for the same hour and activity level is what makes a resting tachycardia visible while exercise stays normal.

The alert asks a question. Treating the crew as a sensor with a voice turns a false positive into training data, and turns a true positive into a conversation that ends in a protocol rather than a red box.
Safety is in code, not in the prompt. The language model writes the explanation; steps, timers, red flags and drug constraints come from the knowledge base and the pharmacy database. That is why the assistant still works with no model at all.
The delay is real. Time is a first-class quantity in the engine: the flight surgeon's view is the ship as it was nineteen minutes ago, because the API serves delayed data rather than an apology banner.

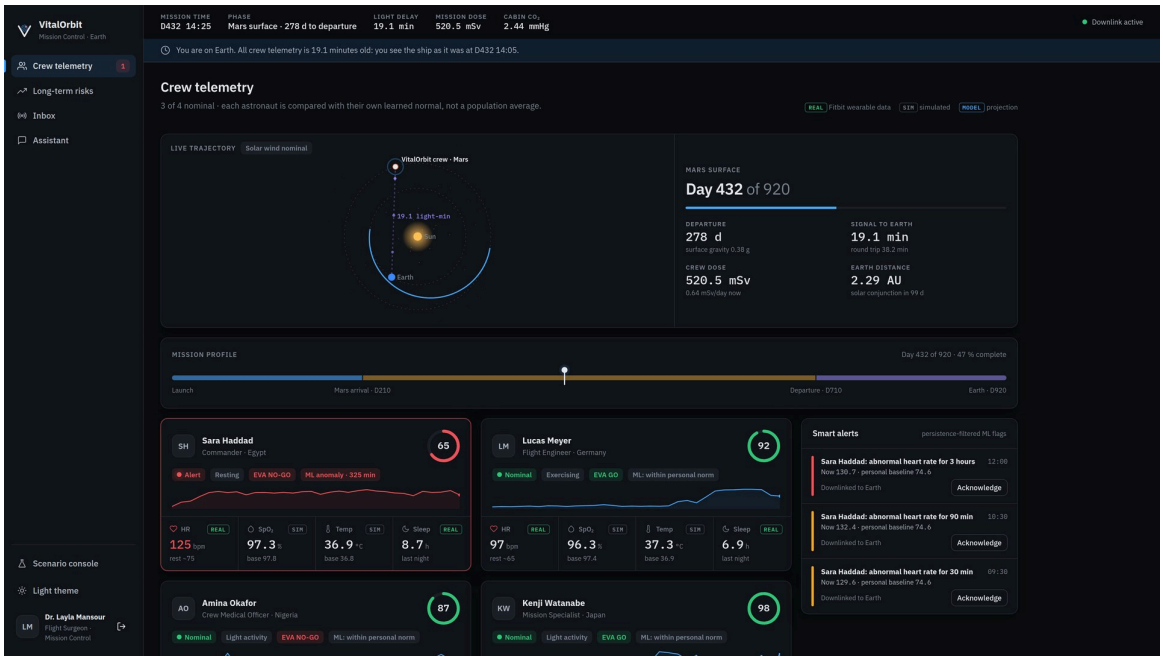


Figure 6: The same system seen from Mission Control. The flight surgeon sees everything, nineteen minutes late, because the delay is applied to the data rather than drawn on the screen: the banner states which instant of the ship is being displayed.

8 Reproducing the demo

```
git clone https://github.com/3AMURR/VitalOrbit
cd VitalOrbit && ./run.sh
open http://localhost:8000
```

No API key is required: with neither Claude nor a local model configured, the assistant answers from onboard protocols. Optional local intelligence: `ollama pull qwen2.5:1.5b`, which the app detects automatically.

Login	PIN	What it demonstrates
Amina Okafor, CMO	3333	Whole crew, alerts, risk board, pharmacy management
Sara Haddad, Commander	1111	An astronaut's own record, the check-in prompt, the assistant
Dr. Layla Mansour, Flight Surgeon	9999	Mission Control on Earth, seeing everything 19 minutes late

The two-minute demonstration: sign in as the CMO and read the trajectory; inject tachycardia on Sara from the scenario console and watch the card turn amber, then red, as alerts escalate; sign in as Sara and answer the check-in with “I don’t feel well” so the assistant opens on the cardiac protocol with the drug checked against the pharmacy; send the case report to Earth; sign in as the flight surgeon and receive it nineteen minutes late; open the risk board and find Kenji’s SANS and the radiation projection crossing 600 mSv; finish in the pharmacy, where a medicine valid by its label has expired under radiation.

9 Limitations

We would rather state these than have them found.

The vitals are real only where a wrist wearable measures them. SpO2 and temperature are simulated with physiological noise, so the detection results measure the method, not the fidelity of a sensor we did not have. The risk projections inherit published rates but our own mitigation factors; they are directional, not clinical. The assistant gives first-aid guidance and always escalates; it is not a diagnosis. The pharmacy’s per-drug radiation sensitivities are assumptions informed by one body of literature, not measured values. Nothing here has been validated with crew, and the demo runs on a laptop rather than flight hardware, where the same code would need certification, redundancy and a watchdog.

Next, in order: a second dataset for cross-person validation, a real SANS imaging pipeline, and a hardware-in-the-loop run on a radiation-tolerant board.

10 Team, links and references

Team CUPIQ, Egyptian Chinese University: Maroo built everything during the 48-hour NASA Space Apps Cairo 2026 hackathon, with help from Claude (Anthropic).

Repository	github.com/3AMURR/VitalOrbit
30-second video	public link, added at submission
Presentation video	public link, added at submission
This document	VitalOrbit_submission_document.pdf

References. Furberg et al., Crowd-sourced Fitbit datasets 03.12.2016–05.12.2016, Zenodo 53894, CC-BY 4.0.
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